

Suicidal Thought Detection Using NLP(Natural Language Processing) on Reddit Data

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Abstract—Our study harnesses the power of NLP to contribute to the identification of suicidal ideation in text data. In the realm of mental health, identification of suicidal ideation at an early stage is of paramount importance for suicide prevention efforts. This paper presents a novel approach to suicide ideation detection using Natural Language Processing (NLP) techniques. Faced with a dearth of publicly available datasets for this critical task, we have contributed a valuable resource by curating a dataset from the "SuicideWatch" and "depression" subreddits on the Reddit platform, collected via the Pushshift API. Specifically, we employ the LSTM and a Random Forest classifier separately to achieve promising results in this vital area of research. This work not only advances the field of NLP-based suicide ideation detection but also contributes a valuable dataset for future investigations, potentially saving time and resources for researchers and professionals dedicated to the prevention of suicide and the improvement of mental health. We could achieve up to 93% accuracy in suicidal thought analysis using NLP techniques.

Index Terms—NLP, Reddit, Pushshift, API, LSTM, Random Forest Classifier.

I. INTRODUCTION

In the contemporary digital landscape, social networking sites have emerged as global agora, reshaping communication dynamics and offering unprecedented insights into human emotions [1]–[4]. Among these platforms, Reddit stands out as a rich source for sentiment research due to its diverse subreddits, fostering candid discussions, including those related to mental health [5]. Addressing the pressing issue of suicide, this research employs Natural Language Processing (NLP) techniques, specifically leveraging the TF-IDF model, to discern between posts expressing suicidal ideation and non-suicidal sentiments. With over 47,000 lives lost annually to suicide in the United States alone, understanding and preventing suicidal thoughts are critical [6].

This study aims to contribute to mental health research by utilizing Reddit data for sentiment analysis. Section 2 reviews related studies on suicidal thought prediction, while Section

3 outlines the methodology, emphasizing the use of NLP approaches. In Section 4, the paper details the implementation of the TF-IDF model. Section 5 discusses results, highlighting the significance of NLP in addressing mental health challenges through social media analysis. In conclusion (Section 6), the research underscores the value of this approach for suicide prevention initiatives in the digital age.

II. RELATED WORKS

In [7], TFIDF and BoW are used in tandem to precisely differentiate between positive and negative tweets. They discovered that by utilizing the TF-IDF vectorizer, the precision of sentiment analysis can be greatly improved and simulation results demonstrate the efficiency of our suggested method. Using the NLP approach, we achieved 85.25% accuracy in sentiment analysis. In [8], The author provided an automated conversational platform that was utilized to identify depression-related risks as a preliminary strategy. The platform was designed to interpret conversations through NLP and machine learning. The suggested two-phased platform's initial phase would examine discussion and sort related emotions into four categories: "happy," "neutral," "depressive," and "suicidal." In [9], The authors suggested a categorization strategy, deep neural networks, Bi-LSTM, CNN, and self-attention are used in this model. , which they demonstrated on several datasets. Furthermore, they contrast three pre-trained word-embeddings for word encoding. The optimistic findings achieved on cutting-edge datasets allow us to test the model's validity and examine the optimum word embeddings to use for emotion identification. They suggest their model as a starting point for further research in the issue because deep learning is so significant in the academic world.

In [10], the authors provided the findings of a thorough mapping research to arrange the available published information. We looked for studies completed between 2015 and 2020 in electronic research databases of the scientific literature using a

stepwise PRISMA approach. They uncovered 92 relevant studies on sentiment analysis of student input in learning platform environments out of 612 initially discovered. The mapping results revealed that, despite the mentioned limitations, the area is rapidly expanding, particularly in terms of the use of DL, which is the most current trend. They outlined some elements that must be considered in order to contribute to the maturity of research and development in the area.

In [11], They demonstrated that using social media data to identify people at risk of suicide is viable. They used NLP and ML (specifically, deep learning) techniques that described proposals for an automated method for predicting suicide risk that may be utilized by persons without specialist mental health training (for example, a primary care practitioner). They also discussed the ethical implications of such technology and the implications for privacy. This technique was only utilized for those who "opted in" for the analysis and intervention, but it permitted scalable screening for suicide risk, potentially identifying many people at risk before they contacted a health care system.

This presented a significant cultural concern regarding the trade-off between privacy and prevention—they have the technology that has the potential to save lives, but it only reached a small percentage of the population who could be at risk due to respect for their privacy.

In [12], They tested weakly supervised algorithms for detecting "current" events. Suicidal ideation in electronic health record (EHR) systems was derived from unstructured clinical notes. Weakly supervised machine learning approaches use imprecise labels for training, which reduces the effort of producing a big dataset.

The dataset had been manually annotated. After identifying a cohort of 600 patients at risk for suicidal ideation, they used a rule-based NLP technique to categorize the training and validation notes ($n = 600$). Using this vast collection of clinical observations, they developed multiple ML statistical models, including a logistic classifier, support vector machines (SVM), and a Naive Bayes classifier, as well as one deep learning model, a text classification convolutional neural network (CNN), to be tested on a hand reviewed test set ($n = 837$).

Our research is pulsing not merely with technological prowess, but also with the tremendous potential to save lives and alleviate suffering. We have given the scientific community a rich dataset, a tapestry of human emotions that calls out to other researchers who share our dedication to the improvement of mental health and the prevention of suicide. These findings, far from being simple statistics, reflect the unspoken cries for assistance of people in need, and it is our serious duty to listen and respond.

III. PROPOSED METHODOLOGY

This section summarizes our ideas and focuses on a method for performing sentiment analysis on Reddit data.

Figure 1 gives an architectural breakdown of the entire process and outlines the methods used in this study to identify suicidal thoughts. The underlying principle of the created

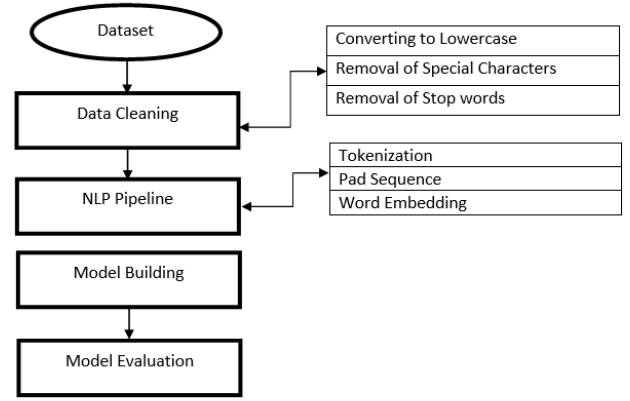


Fig. 1. Architecture Overview of Proposed Methodology

method is the use of text-type data derived from the Reddit site. The process consists of a number of significant steps, each of which contributes to the thorough examination of Reddit data to find suicidal ideation.

Algorithm 1

Step 1: Data Preparation

1. Import necessary libraries:
 - Import the required libraries such as WordCloud, numpy, pandas, sklearn, neattext, matplotlib, plotly, keras, tensorflow, and others.
2. Load the dataset:
 - Load the dataset, which presumably contains text data along with labels indicating whether the text expresses suicidal thoughts or not.

Step 2: Data Analysis and Visualization

3. Explore the data:
 - Examine the distribution of classes in the dataset, i.e., "Suicide" and "Not Suicide."
4. Visualize the class distribution:
 - Create bar charts and pie charts to visualize the distribution of suicide-related and non-suicidal content.

Step 3: Text Preprocessing

5. Clean the text data:
 - Lowercase the text.
 - Remove special characters.
 - Remove stopwords.

Step 4: Tokenization and Word Frequency

6. Tokenize the cleaned text:
 - Tokenize the preprocessed text data.
7. Calculate word frequency:
 - Create a word frequency DataFrame to analyze word frequencies.

Step 5: Word Cloud Visualization

8. Generate a word cloud:
- Generate a word cloud visualization to gain insights into the most frequent words in the dataset.

Step 6: Data Splitting and Label Encoding

9. Split the data:
 - Split the data into training and testing sets.
10. Encode labels:
 - Encode the class labels ("Suicide" and "Not Suicide") using LabelEncoder.

Step 7: Word Embedding with Pretrained Vectors

11. Load pretrained word embeddings:
 - Load pretrained word embeddings (GloVe) to create an embedding matrix for words in the dataset.

Step 8: Model Building and Training

12. Define the model architecture:
 - Create a Sequential model.
 - Add an Embedding layer with pretrained word embeddings.
 - Add an LSTM layer with return sequences.
 - Add a GlobalMaxPooling1D layer.
 - Add Dense layers with ReLU activation.
 - Compile the model using binary cross-entropy loss and SGD optimizer.
13. Train the model:
 - Train the model on the training data, monitoring validation performance.
 - Utilize early stopping and learning rate reduction callbacks.

Step 9: Model Evaluation and Visualization

14. Visualize training history:
 - Plot accuracy curves for training and validation data.

Step 10: Model Saving

15. Keep the trained model:
 - Keep the learned model for further use.

Step 11: Suicidal Thought Detection Function

16. Create a function to detect suicidal thoughts:
 - Input: Text data.
 - Preprocess the input text.
 - Tokenize, pad, and process the text for prediction.
 - Use the trained model to predict whether the text contains suicidal thoughts (based on a threshold).
17. Call the detectSuicide function with sample text data for detection.

- Remove the 'Unnamed: 0' column if present.
- Check and print basic information about the dataset.

Step 4: Data Visualization

- Visualize the class distribution using a countplot.

Step 5: Text Length Analysis

- Calculate the length (in terms of words) of each text.
- Analyze the text length statistics, including quantiles.

Step 6: Filter Texts Based on Length

- Remove texts with a length exceeding a certain threshold (317 words in this case).
- Visualize the class distribution after filtering.

Step 7: Word Frequency Analysis

- Tokenize and count the frequency of words in the text data.
- Filter out rare and common words based on quantiles (0.99% and 99.9% quantiles).
- Visualize word frequency statistics.

Step 8: Feature Engineering

- Create binary features for selected frequent words in the text data.

Step 9: Label Encoding

- Encode the 'class' column, converting 'suicide' to 1 and 'non-suicidal' to 0.

Step 10: Data Visualization (Optional)

- Visualize the relationship between the 'class' and the length of texts.

Step 11: Feature Correlation Analysis

- Calculate the Pearson correlation coefficient between binary features and the target 'class'.

Step 12: Model Training and Evaluation

- Split the data into training and test sets (1% for training).
- Initialize a Random Forest Classifier.
- Perform hyperparameter tuning using GridSearchCV.
- Train the Random Forest model on the training data.
- Predict using both training and test data.
- Evaluate the model using accuracy and recall scores.

The core of our study is the proposed technique, which directs our efforts to find and comprehend the existence of suicidal ideas in the Reddit dataset. The findings and their implications will be presented in the next sections of this essay as they relate to studies on mental health and suicide prevention.

IV. IMPLEMENTATION

A. Data Collection

The first step entails gathering textual information from Reddit, concentrating on postings pertinent to suicidal and non-suicidal thoughts. This information, which includes posts from the "SuicideWatch" and "depression" subreddits as well as non-suicidal messages from r/teenagers, was gathered using the Pushshift API.

Algorithm 2

Step 1: Import Libraries

- Import necessary libraries including NumPy, pandas, matplotlib, seaborn, warnings, and scikit-learn.

Step 2: Load Data

- Load the dataset from the specified file path using Pandas.

Step 3: Data Preprocessing

	Unnamed: 0	text	class
0	2	Ex Wife Threatening SuicideRecently I left my ...	suicide
1	3	Am I weird I don't get affected by compliments...	non-suicide
2	4	Finally 2020 is almost over... So I can never ...	non-suicide
3	8	i need helpjust help me im crying so hard	suicide
4	9	I'm so lostHello, my name is Adam (16) and I've...	suicide

Fig. 2. Dataset Used for Suicidal Thought Detection

B. Data pre-processing

1) *Data Extraction*: The collected data is extracted from the API and organized for further analysis.

2) *Removal of E-mails Converting to Lower-Case*: Firstly, the text dataset underwent preprocessing where emails were removed, and the entire dataset was converted to lowercase.

3) *Tokenization*: Text data is tokenized into individual words or tokens to facilitate subsequent processing.

'cousin asleep leg im scared dont want wake dont know'

Fig. 3. Tokenized Data

4) *Lemmatization*: Text lemmatization techniques are applied to reduce words to their root forms, aiding in the removal of variations and enhancing feature extraction.

5) *Stopwords Removal*: Common stopwords, such as "and," "the," and "in," are removed to focus on meaningful content.

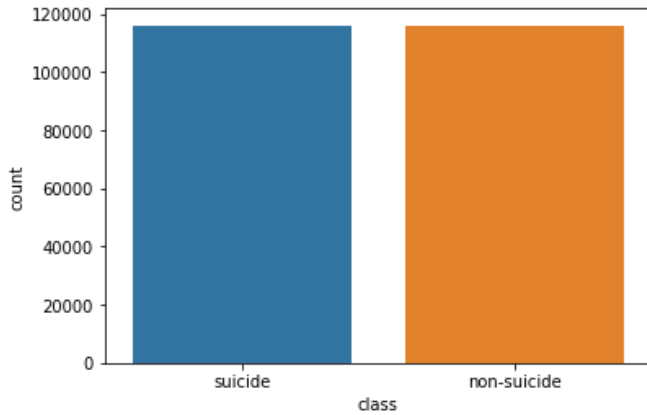


Fig. 4. Data Visualization

C. Feature Extraction

In a departure from the conventional TF-IDF method, our feature extraction process employed a custom approach to enhance the interpretability and relevance of features. Specifically, we crafted binary features based on the frequency of selected words within the text data. This method involved encoding the 'class' column, assigning 'suicide' a numerical value of 1 and 'non-suicidal' a value of 0. Subsequently, we calculated the Pearson correlation coefficient between the

binary features and the target 'class'. This innovative feature extraction technique aimed to capture the distinctive patterns associated with suicidal and non-suicidal posts, providing a nuanced representation of the dataset. The binary features, generated through this process, were then utilized for training and evaluating machine learning models. This tailored feature extraction methodology holds the potential to reveal intricate relationships between specific word occurrences and the classification of posts, fostering a more insightful understanding of the dataset and facilitating improved model performance.

	Word	Frequency
4	I	890895
18	TO	558762
16	AND	464789
90	THE	350903
30	A	339305
...	...	...
404	CAN'T	35968
296	HAD	35618
460	DON'T	35336
1328	I'VE	35261
158	ONLY	34925

70 rows × 2 columns

Fig. 5. Word Frequency

This model determines a word's significance in relation to the entirety of the dataset.

1) *Word Embedding*: For deep learning models like LSTM and Bi-LSTM, we utilized GloVe and Word2Vec word embeddings. GloVe captures global co-occurrence statistics, offering insights into semantic relationships, while Word2Vec focuses on local context, enhancing syntactic understanding. This dual embedding approach enriches our models with nuanced representations, facilitating improved performance in tasks like sentiment analysis and post-classification.

D. Algorithm Utilization

Unique algorithms and software packages are designed expressly for the purpose of helping each step of the process at hand. Stemming, lemmatization, and stopwords elimination are just a few of the NLP operations that are performed using the Natural Language Toolkit (NLTK).

	class	Length	SOME	ME	FILLER	WE	HAS	FUCKING	WHY	BENIG	...	WEIRD	WHAT'S	LONGER	OTHERS	THANK	SUPPOSED	HEAR	FRIENDS	MOTH
0	suicide	143	0	0	0	0	1	0	0	0	...	0	0	0	0	0	1	0	0	0
1	non-suicide	27	0	0	0	0	0	0	0	0	...	1	0	0	0	0	0	0	0	0
2	non-suicide	26	0	0	0	0	1	1	0	0	...	0	0	0	0	0	0	1	0	0
3	suicide	9	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
5	suicide	289	0	1	0	0	0	0	0	1	...	0	0	0	0	0	0	0	0	0

Fig. 6. Separate Variables for Words Based on Their Presence

E. Classifier Model

In this study, we investigate the performance of the Random Forest classifier and LSTM (Long Short-Term Memory) neural networks as two different classifier models for detecting suicidal thoughts. We have also applied Bi-lstm in this research.

V. RESULTS & DISCUSSION

This section contains the results of our experimentation and goes into the analysis of the findings obtained by using LSTM, Bi-LSTM and Random Forest Classifier models to detect suicidal ideation in text data. We will first present a brief assessment of the model's efficiency in terms of accuracy, precision, recall, and F1 score, followed by a discussion of the ramifications of these findings.

The Bi-LSTM model performs admirably overall, with an accuracy of 92.88%. This statistic represents the proportion of correctly classified cases in the test dataset. The model's precision score of 95.02% demonstrates its ability to correctly classify postings as expressing suicidal ideation while reducing false positives. A recall score of 90.50% demonstrates the model's ability to recognize true suicidal ideation messages, reducing false negatives. The F1 score of 92.71% strikes a good balance between precision and recall, indicating a solid and well-rounded model. The Bi-LSTM model's exceptional performance is due to its capacity to recognize sequential dependencies within textual input. The model effectively evaluates the context and complex patterns in the text by leveraging Bidirectional Long Short-Term Memory (LSTM) units, allowing for accurate predictions.

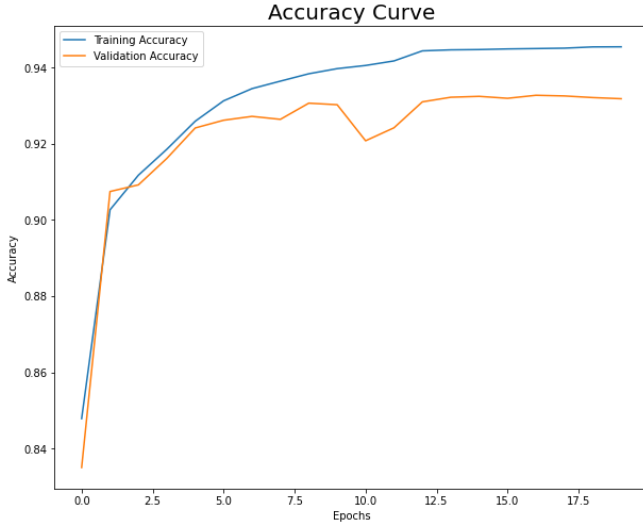


Fig. 7. Accuracy Curve

While the Random Forest Classifier achieves a decent accuracy of 83.75%, there are significant discrepancies when compared to the Bi-LSTM model. The precision score of 86.15% indicates that the classifier has a moderately strong ability to reduce erroneous positives. The recall score of 76.73%, on the other hand, indicates a modest ability to detect true suicidal ideation posts, resulting in a slightly lower F1 score of 81.17%.

The Random Forest Classifier works on the ensemble learning concept, combining predictions from numerous decision trees. While it is useful for a variety of tasks, its efficacy

in identifying complicated sequential relationships within text data may be limited when compared to Bi-LSTM. This could account for the differences in recall and overall F1 score.

The performance metrics obtained for the Bi-LSTM, LSTM and Random Forest Classifier models are summarized in Table 1 below-

TABLE I
COMPARISON BETWEEN PARAMETERS

Model	Accuracy	Precision	Recall	F1 Score
Bi-LSTM	92.88%	95.02%	90.50%	92.71%
LSTM	92.62%	94.73%	90.26%	92.44%
Random Forest Classifier	83.75%	86.15%	76.73%	81.17%

The table 2 outlines various feature extraction, machine learning, and embedding techniques along with deep learning algorithms applied in different studies. The first entry employs TF-IDF for feature extraction, SVM for machine learning, and Word2Vec for embedding, employing LSTM and CNN in deep learning, achieving a notable accuracy of 90.3% as reported in [5]. The second study utilizes TF-IDF, LIWC and Sentiments for feature extraction, employing a range of machine learning algorithms such as RF, SVM, LR, and ZeroR, resulting in a high accuracy of 92%, as documented in [13]. Another approach combines TF-IDF, N-Gram, and LIWC for feature extraction with various machine learning algorithms, reaching an accuracy of 73.6%, as reported in [9]. Finally, the current research employs binary feature correlation for feature extraction, RF for machine learning, and Word2Vec and Glove for embedding, with Bi-LSTM as the deep learning algorithm, achieving an impressive accuracy of 93%, as presented in this paper.

The findings highlight the importance of NLP approaches, particularly LSTM-based models, in detecting suicidal ideation in textual data. The LSTM model's high accuracy, precision, and recall reveal its ability to detect minor verbal clues suggestive of suicidal ideation.

Future studies could look into combining multiple NLP models, creating hybrid models, or including domain-specific features to improve the efficacy of suicide ideation detection systems. Efforts to reduce the model's false positives and false negatives should also be prioritized, as they have substantial ramifications in real-world applications. Finally, our research highlights the significant prospects of NLP approaches, particularly LSTM models, in the detection of suicidal ideation. While the Random Forest Classifier produces decent results, the capacity of the LSTM model to catch intricate textual patterns is a significant leap in the field of mental health research and suicide prevention initiatives.

VI. CONCLUSIONS

In a world dominated by the silent battle of countless people dealing with suicide ideation, our study shines as a beacon of hope and creativity. Our path was distinguished by unwavering determination, driven by a desire to apply NLP to detect, comprehend and assist people in need. The

TABLE II
COMPARISON WITH PREVIOUS WORKS

Feature Extraction Techniques	Machine Learning Algorithms	Embedding Techniques	Deep Learning Algorithms	Best Performing Model	Metric & Result	Ref
TF-IDF	SVM	Word2Vec	LSTM, CNN	LSTM-Attention CNN	90.3%	[5]
TF-IDF, LIWC, Sentiment	RF, SVM, LR, ZeroR	NA	NA	SVM	92%	[13]
TF-IDF, N-Gram, LIWC	NB, SVM, KNN, RF	NA	NA	NA	73.6%	[9]
Binary Feature Correlation	RF	Word2Vec, Glove	LSTM, Bi-LSTM	Bi-LSTM	93%	This paper

scope of our investigation immediately highlighted a daunting challenge: the scarcity of publicly available datasets necessary to our attempt. In response, we diligently curated a one-of-a-kind dataset culled from the candid and heartfelt expressions published on the Reddit platform's "SuicideWatch" and "depression" subreddits. This dataset, which spans over a decade of human emotions, is more than just a resource; it demonstrates our persistent commitment to the goals of suicide prevention and mental health promotion. Our research was built around a variety of Natural Language Processing approaches. We set out on a journey to grasp the complexities of human language, the nuances of emotion, and the terrible reality of suicidal ideation. Our method avoided the one-size-fits-all philosophy, instead employing a hybrid of two powerful models: the LSTM (Long Short-Term Memory), Bi-LSTM and the Random Forest Classifier. We produced exceptional results through a dynamic integration of machine learning algorithms. In sentiment analysis, the Bi-LSTM model, which is known for its ability to capture sequential patterns and contextual information, achieved an extraordinary accuracy rate of up to 93%. The Random Forest Classifier, on the other hand, produced equally encouraging results, with a Test Accuracy of 83.7%. Finally, this thesis represents more than just a collection of studies; it represents the indomitable spirit of human compassion and inventiveness. It demonstrates the revolutionary power of NLP approaches, as demonstrated by the LSTM and Random Forest Classifier, in addressing today's most important concerns. As we embark on an unknown future, may our work serve as a spark for greater inquiry, collaboration, and the creation of enhanced suicide prevention measures.

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